

CARLOS BLANCO

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EXPERIENCE

Software Engineering Intern — *Snow College* — Feb 2025 – May 2025

- Modernized a 4-year-old legacy bacterial-identification web app, cutting page-load time and eliminating recurring deployment bugs, by migrating the stack to ASP.NET, React, Redux, and PostgreSQL.
- Enabled zero-downtime deployments by architecting a microservices system with Docker Compose for local development and Kubernetes in production.
- Increased deployment speed and reliability for a tool used in biology classes by 50+ students each term, by building a GitHub Actions CI/CD pipeline targeting college on-premise servers.

Full-Stack Developer Intern — *Sanpete Pantry* — May 2024 – Aug 2024

- Increased donations by ~108% and user engagement by ~22% for a nonprofit serving 3,340+ residents, by designing and launching a new ASP.NET / JavaScript / Tailwind CSS web application from scratch.
- Enabled secure online payments, including support for scheduled payments (monthly, weekly, etc) by integrating Neon CRM.

PROJECTS

WorkspaceAI — workspaceai.carlosblancodev.com

- Implemented a RESTful ASP.NET API with JWT/OAuth2 authentication, ownership-based authorization, and custom validation attributes for request sanitization, using EF Core to persist data to PostgreSQL.
- Cut bundle transfer size ~57% through lazy loading in a responsive Angular front end with standalone components, Signals, route guards, and reactive forms with custom validation.
- Built a Python FastAPI document-processing server that extracts text from PDF, DOCX, and TXT files (PyPDF, python-docx) for AI-driven document analysis.
- Self-hosted an Ollama instance running Llama 3.2 (1B) for document summarization, eliminating recurring API costs and preserving user privacy.
- Migrated Ollama instance to Oracle Cloud Infrastructure VM (Standard.A1.Flex), reducing response latency by ~18%.
- Implemented request authentication using Caddy reverse-proxy to secure API access for self-hosted model.
- Built real-time chat functionality with a SignalR (WebSocket) hub in ASP.NET, achieving an average response time of ~4.6s between the Ollama instance and the user.
- Built a Python RAG pipeline that includes text normalization, chunking, embedding, vector persistence using pgVector, as well as cosine-similarity retrieval, to optimize LLM context window usage for document Q&A.
- Enabled fast, secure document upload, download, preview, and deletion of user documents by integrating Cloudflare R2 Object Storage with pre-signed URLs exposed through the ASP.NET API.

S&P 500 Model and Web App — sp500.carlosblancodev.com

- Achieved 57.41% directional accuracy predicting the S&P 500's closing value, outperforming a 50% random baseline, by designing, training, and evaluating a Random Forest model in Python on Yahoo Finance data.
- Built a React (TypeScript, Vite) SPA to visualize 10+ years of market data, rolling averages, tech stock trends, and economic-uncertainty indicators.
- Served real-time ML predictions with a low-latency FastAPI service that loads the trained model and fetches data efficiently.
- Reduced end-to-end API latency ~95% (from ~3.0s to ~130ms), by implementing Redis caching.
- Cut average deploy time to 2m 26s, by architecting an automated CI/CD pipeline (GitHub Actions) to a Kubernetes home-lab cluster with public access secured via Cloudflare Tunnel.

EDUCATION

M.S. in Data Science — *Johns Hopkins University*

Aug 2025 – Present

- GPA: 4.0 / 4.0

B.S. in Software Engineering — *Snow College*

Aug 2023 – May 2025

- GPA: 3.97 / 4.0

SKILLS

Languages: C#, JavaScript, TypeScript, Python, SQL, C++, R, HTML, CSS.

Frameworks & Libraries: ASP.NET Core, Angular, React, FastAPI, EF Core, Tailwind CSS, AutoMapper, Redux.

Cloud & DevOps: Docker, Kubernetes, GitHub Actions, Cloudflare (Tunnel, R2), Azure, AWS, Git.